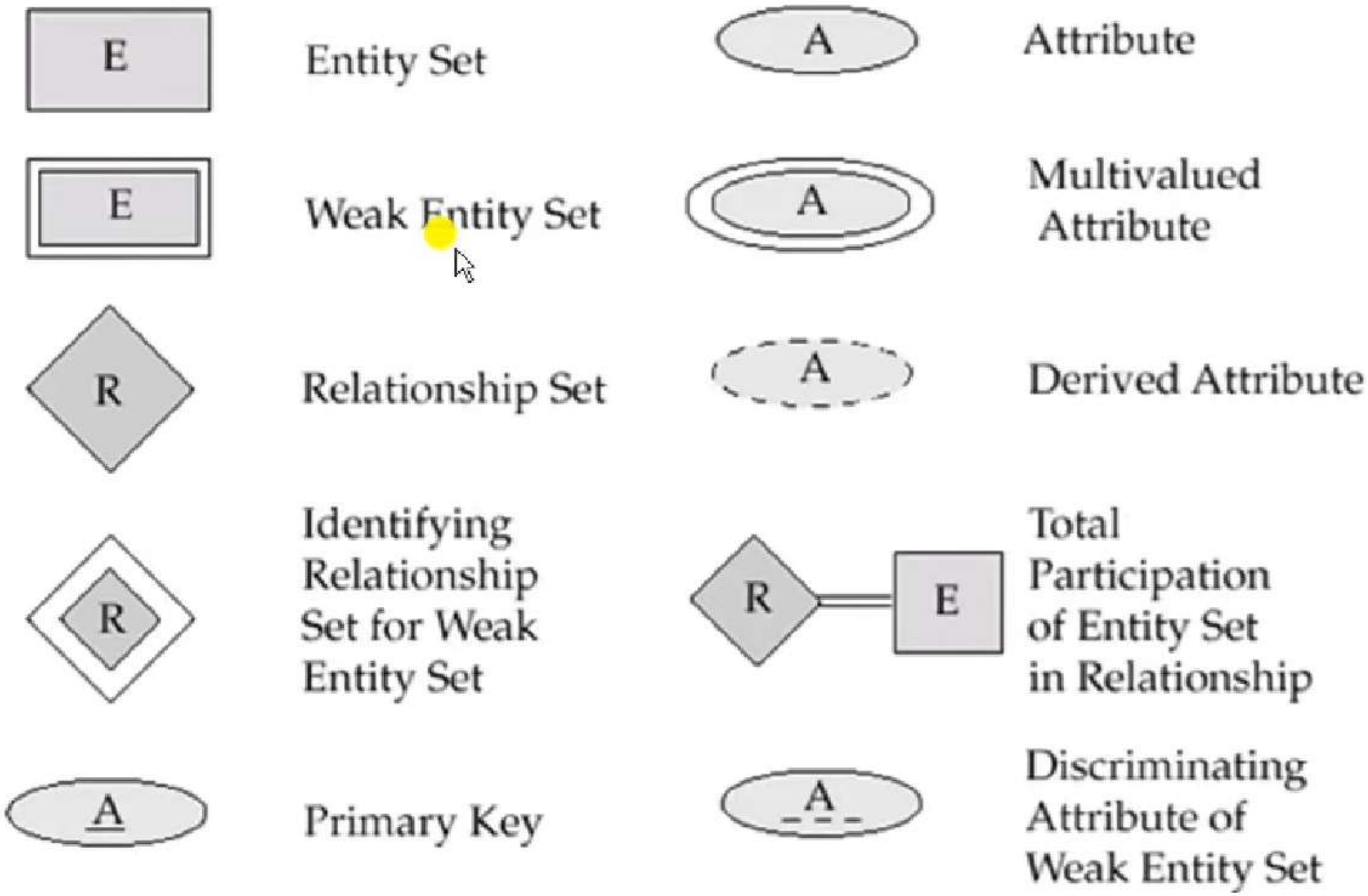
Steps to draw an ER Diagram

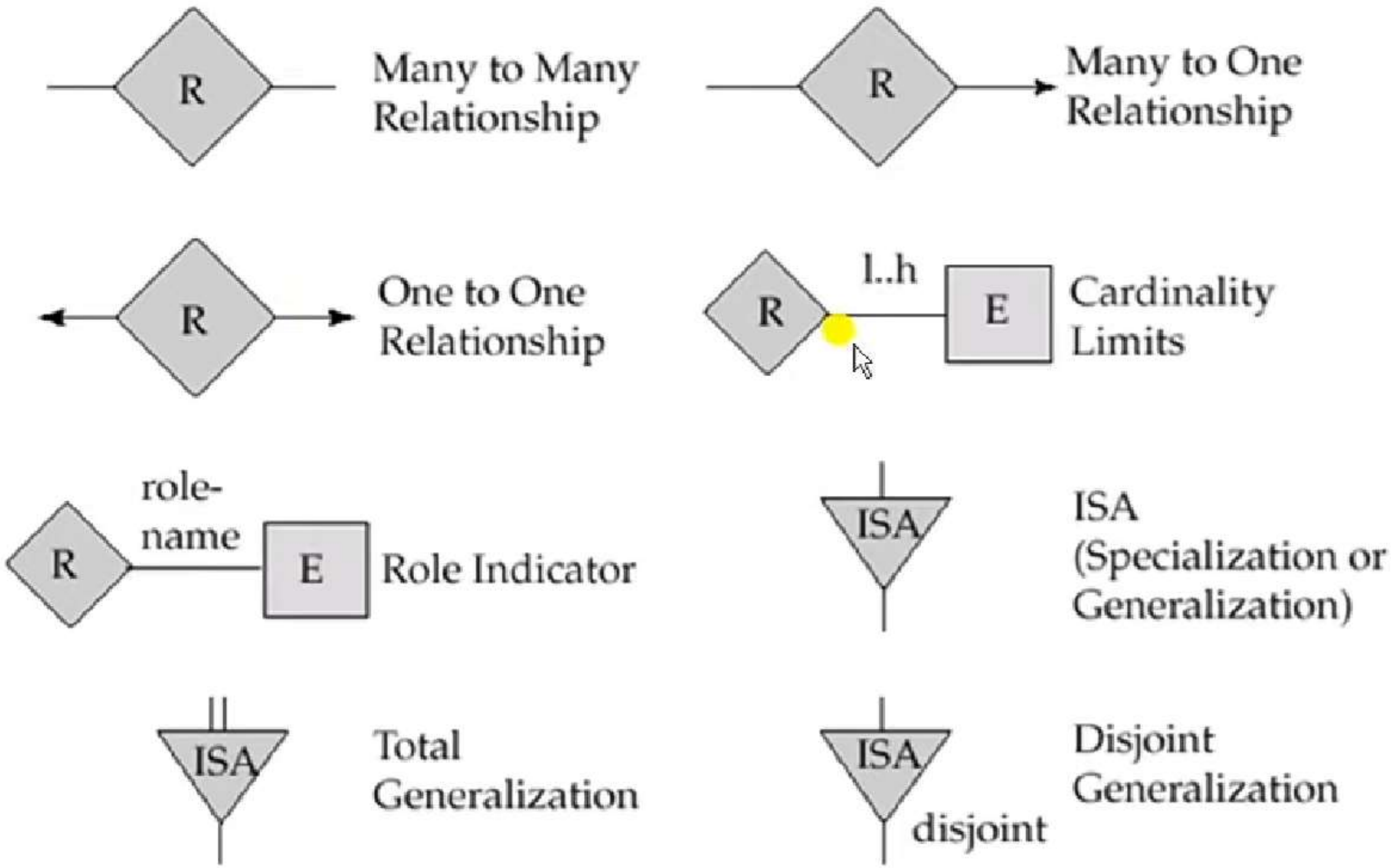
1. Identify the **entities**
2. Identify the **attributes**
3. Identify the **primary key**
4. Identify the **relationships and participation constraints**

Symbols Used in E-R Notation

SHANU KUTTAN CSE CLASSES

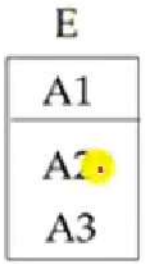


Symbols Used in E-R Notation



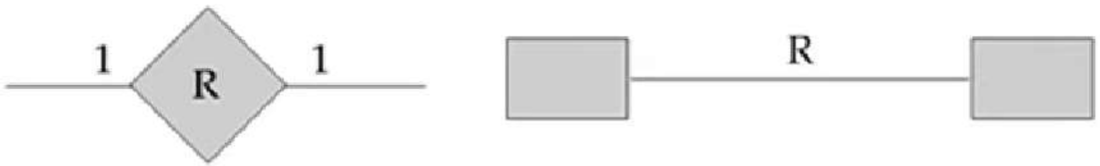
Alternative E-R Notations

Entity set E with attributes A1, A2, A3 and primary key A1



Many to Many Relationship

One to One Relationship



Many to One Relationship

Requirements for a Bank Database:

- The bank is organized into branches. Each branch is located in a particular city and is identified by a unique name. The bank monitors the assets of each branch.
- Bank customers are identified by their customer id values. The bank stores each customer's name and the street and city where the customer lives. Customers may have accounts and can take out loans. A customer may be associated with a particular bank employee, who may act as a loan officer or personal banker for that customer.
- Bank employees are identified by their employee id values. The bank administration stores the name and telephone number of each employee, the names of the employee's dependents, and the employee id number of the employee's manager. The bank also keeps the track of the employee's start date and, thus, length of employment.
- The bank offers two types of accounts - savings and checking accounts. Accounts can be held by more than one customer, and a customer can have more than one account. Each account is assigned a unique account number. The bank maintains a record of account's balance and the most recent date on which the account was accessed by each customer holding the account. In addition, each savings account has an interest rate and overdrafts are recorded for each checking account.
- A loan originates at a particular branch and can be held by one or more customers. A loan is identified by a unique loan number. For each loan, the bank keeps track of the loan amount and the loan payments. Although a loan payment number does not uniquely identify a particular payment among those for all the bank's loans, a payment number does identify a particular payment for a specific loan. The date and amount are recorded for each payment.

1. Identifying the ENTITIES

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1. Identifying the ENTITIES

Banking Database consists of 6 entities:

1. Branch
2. Customer
3. Employee
4. Account
5. Loan
6. Payment (it is a weak entity depends on Loan entity)

2. Identify the ATTRIBUTES

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2. Identify the ATTRIBUTES

1. **branch**: <u>branch-name</u>, branch-city, assets
2. **customer**: <u>customer-id</u>, customer-name, customer-street, customer-city
3. **employee**: <u>employee-id</u>, employee-name, telephone-number, start-date, dependent-name, employment-length
4. **account**: <u>account-number</u>, balance
5. **loan**: <u>loan-number</u>, amount
6. **payment**: <u>payment-number</u>, payment-date, payment-amount

3. Identify the PRIMARY KEY

Entity	Primary Key
branch	branch-name
customer	customer-id
employee	employee-id
account	account-number
loan	loan-number
payment	payment-number

4. Identify the RELATIONSHIPS

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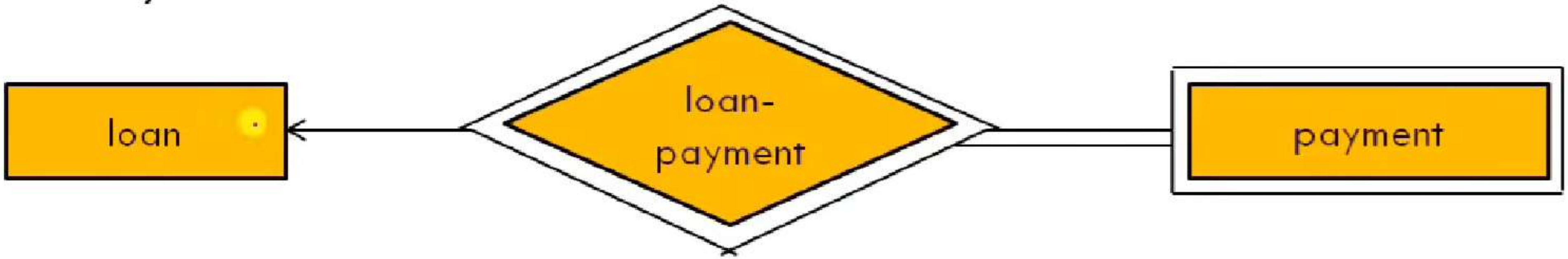
4. Identify RELATIONSHIP & PARTICIPATION CONTR T

- A **customer** can have multiple **accounts** and an account can be held by multiple customers. Cardinality will be M : N

- **Customers** are allowed to take **loans** from the bank, a loan can be held by one or more customers. Cardinality will be M : N

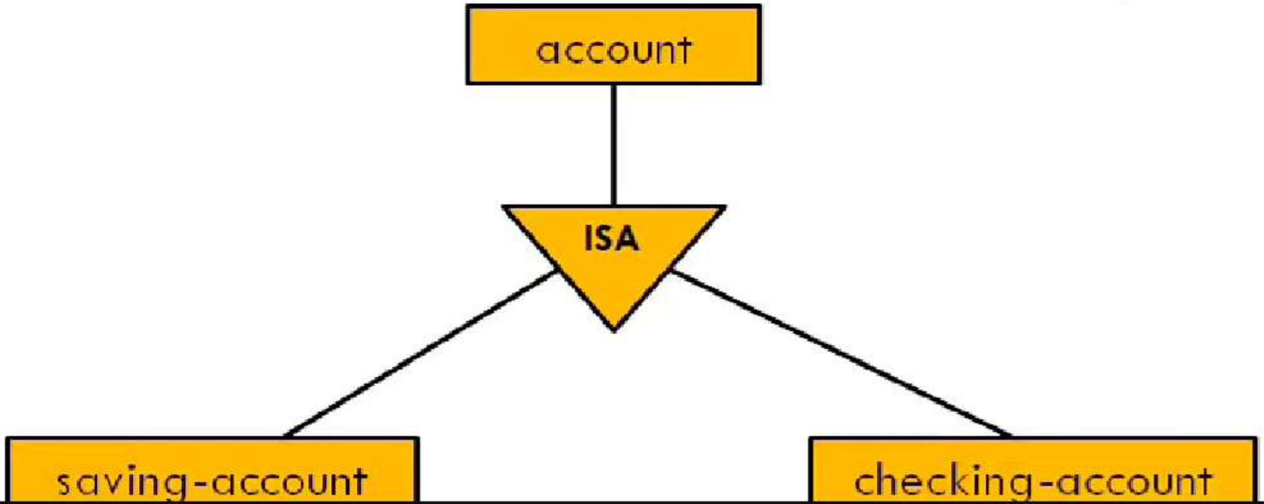
- Each **loan** is paid back with the multiple **payments** having payment number. Cardinality will be 1 : N



- A **customer** can be an **employee** of the bank. Cardinality will be M : N

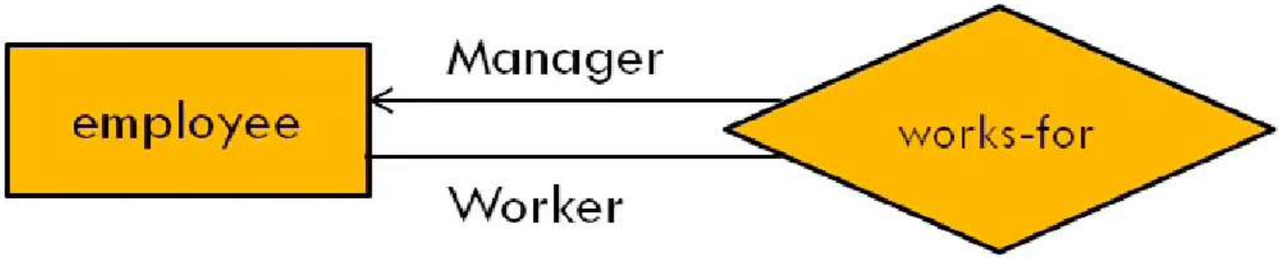
- A **branch** can have multiple **accounts.** Cardinality will be 1 : N

- An **account** can be divided into two accounts: saving account and checking account



- A **branch** of bank give **loans** to the customers. Cardinality will be 1 : N

- **Employees** can be a **manger** or a **worker** of that particular branch. (Recursive Relationship exist)



5. Construct ER Diagram

ER Diagram Of Banking Enterprise